MELODY / 32 BARS

01 / DATA MODEL

```
type Note = { pitch: string; beats: number };
type Bar = {
  meter: "4/4" | "3/4" | "5/4";
  chord: string;
  harmony: string[];
  bassRoot: string;
  bassFifth: string;
  melody: Note[];
};

const n = (pitch: string, beats: number): Note => ({
    pitch, beats });
const bars: Bar[] = [
```

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02 / BARS 01-04

```
{ meter:"4/4", chord:"C",  harmony:["C4","E4","G4"],
    bassRoot:"C2", bassFifth:"G2",
    melody:[n("C4",1),n("D4",.5),n("E4",.5),n("D4",2)] },

{ meter:"4/4", chord:"Am",
    harmony:["A3","C4","E4"],
    bassRoot:"A1", bassFifth:"E2",
    melody:[n("C4",1),n("A3",1),n("C4",1),n("E4",1)] },

{ meter:"4/4", chord:"F",  harmony:["F3","A3","C4"],
    bassRoot:"F1", bassFifth:"C2",
    melody:[n("A4",1),n("G4",.5),n("F4",.5),n("E4",2)] },

{ meter:"4/4", chord:"G",  harmony:["G3","B3","D4"],
    bassRoot:"G1", bassFifth:"D2",
    melody:[n("D4",1),n("G4",1),n("F4",1),n("D4",1)] },
```

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03 / BARS 05-08

```
{ meter:"4/4", chord:"C",  harmony:["C4","E4","G4"],
    bassRoot:"C2", bassFifth:"G2",
melody:[n("C4",.5),n("D4",.5),n("E4",1),n("G4",1),n("E4",1)]
    },

{ meter:"4/4", chord:"Am",
    harmony:["A3","C4","E4"],
    bassRoot:"A1", bassFifth:"E2",
    melody:[n("A4",1),n("G4",.5),n("E4",.5),n("C4",2)] },

{ meter:"3/4", chord:"Dm",
    harmony:["D4","F4","A4"],
    bassRoot:"D2", bassFifth:"A2",
    melody:[n("F4",1),n("A4",1),n("G4",1)] },

{ meter:"3/4", chord:"G",  harmony:["G3","B3","D4"],
    bassRoot:"G1", bassFifth:"D2",
    melody:[n("D4",.5),n("E4",.5),n("F4",1),n("G4",1)] },
```

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04 / BARS 09-12

```
{ meter:"4/4", chord:"Em",
    harmony:["E4","G4","B4"],
    bassRoot:"E2", bassFifth:"B2",
    melody:[n("G4",1),n("B4",1),n("G4",1),n("E4",1)] },

{ meter:"4/4", chord:"Am",
    harmony:["A3","C4","E4"],
    bassRoot:"A1", bassFifth:"E2",
melody:[n("A4",.5),n("C5",.5),n("B4",1),n("A4",1),n("E4",1)]
    },

{ meter:"5/4", chord:"F",  harmony:["F3","A3","C4"],
    bassRoot:"F1", bassFifth:"C2",
melody:[n("F4",1),n("A4",1),n("C5",1),n("A4",1),n("G4",1)]
    },

{ meter:"5/4", chord:"G7",
    harmony:["G3","B3","D4","F4"],
    bassRoot:"G1", bassFifth:"D2",
melody:[n("G4",.5),n("A4",.5),n("B4",1),n("D5",1),n("C5",1),n("
    },
```

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05 / BARS 13-16

```
{ meter:"4/4", chord:"C",  harmony:["C4","E4","G4"],
    bassRoot:"C2", bassFifth:"G2",
    melody:[n("E5",1),n("D5",.5),n("C5",.5),n("G4",2)] },

{ meter:"4/4", chord:"F",  harmony:["F3","A3","C4"],
    bassRoot:"F1", bassFifth:"C2",
    melody:[n("A4",1),n("C5",1),n("A4",1),n("F4",1)] },

{ meter:"4/4", chord:"G7",
    harmony:["G3","B3","D4","F4"],
    bassRoot:"G1", bassFifth:"D2",
melody:[n("G4",.5),n("A4",.5),n("B4",1),n("D5",1),n("B4",1)]
    },

{ meter:"4/4", chord:"C",  harmony:["C4","E4","G4"],
    bassRoot:"C2", bassFifth:"G2",
    melody:[n("C5",4)] },
```

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06 / BARS 17-20

```
{ meter:"4/4", chord:"C",  harmony:["C4","E4","G4"],
    bassRoot:"C2", bassFifth:"G2",
    melody:[n("C5",1),n("D5",.5),n("E5",.5),n("D5",2)] },

{ meter:"4/4", chord:"Em",
    harmony:["E4","G4","B4"],
    bassRoot:"E2", bassFifth:"B2",
    melody:[n("E5",1),n("D5",.5),n("B4",.5),n("G4",2)] },

{ meter:"5/4", chord:"Am",
    harmony:["A3","C4","E4"],
    bassRoot:"A1", bassFifth:"E2",
melody:[n("A4",1),n("C5",1),n("E5",1),n("C5",1),n("B4",1)]
    },

{ meter:"5/4", chord:"D7",
    harmony:["D4","F#4","A4","C5"],
    bassRoot:"D2", bassFifth:"A2",
melody:[n("D5",.5),n("E5",.5),n("F#5",1),n("A5",1),n("G5",1),n(
    },
```

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07 / BARS 21-24

```
{ meter:"4/4", chord:"G",  harmony:["G3","B3","D4"],
    bassRoot:"G1", bassFifth:"D2",
    melody:[n("G5",2),n("D5",1),n("B4",1)] },

{ meter:"4/4", chord:"C",  harmony:["C4","E4","G4"],
    bassRoot:"C2", bassFifth:"G2",
    melody:[n("C5",1),n("E5",1),n("D5",1),n("C5",1)] },

{ meter:"4/4", chord:"D7",
    harmony:["D4","F#4","A4","C5"],
    bassRoot:"D2", bassFifth:"A2",
melody:[n("A4",.5),n("B4",.5),n("C5",1),n("F#5",1),n("E5",1)]
    },

{ meter:"4/4", chord:"G",  harmony:["G3","B3","D4"],
    bassRoot:"G1", bassFifth:"D2",
    melody:[n("D5",1),n("B4",1),n("G4",2)] },
```

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08 / BARS 25-28

```
{ meter:"3/4", chord:"Em",
    harmony:["E4","G4","B4"],
    bassRoot:"E2", bassFifth:"B2",
    melody:[n("E5",1),n("D5",1),n("B4",1)] },

{ meter:"3/4", chord:"Am",
    harmony:["A3","C4","E4"],
    bassRoot:"A1", bassFifth:"E2",
    melody:[n("C5",.5),n("B4",.5),n("A4",2)] },

{ meter:"4/4", chord:"Dm",
    harmony:["D4","F4","A4"],
    bassRoot:"D2", bassFifth:"A2",
    melody:[n("F4",1),n("A4",1),n("C5",1),n("A4",1)] },

{ meter:"4/4", chord:"G7",
    harmony:["G3","B3","D4","F4"],
    bassRoot:"G1", bassFifth:"D2",
melody:[n("G4",.5),n("A4",.5),n("B4",1),n("D5",1),n("F5",1)]
    },
```

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09 / BARS 29-32

```
{ meter:"4/4", chord:"C",  harmony:["C4","E4","G4"],
    bassRoot:"C2", bassFifth:"G2",
    melody:[n("E5",1),n("D5",.5),n("C5",.5),n("G4",2)] },

{ meter:"4/4", chord:"F",  harmony:["F3","A3","C4"],
    bassRoot:"F1", bassFifth:"C2",
    melody:[n("A4",1),n("C5",1),n("A4",1),n("F4",1)] },

{ meter:"4/4", chord:"G7",
    harmony:["G3","B3","D4","F4"],
    bassRoot:"G1", bassFifth:"D2",
melody:[n("G4",.5),n("A4",.5),n("B4",1),n("D5",1),n("B4",1)]
    },

{ meter:"4/4", chord:"C",  harmony:["C4","E4","G4"],
    bassRoot:"C2", bassFifth:"G2",
    melody:[n("C5",4)] },

];
```

MELODY / 32 BARS

10 / EVENTS

```
type Event = { track: "melody" | "chords" | "bass"; pitch:
    string; startBeat: number; beats: number };
const events: Event[] = [];
let startBeat = 0;
for (const [index, bar] of bars.entries()) {
  const barBeats = Number(bar.meter[0]);
  const melodyBeats = bar.melody.reduce((sum, note) => sum
    + note.beats, 0);
  if (melodyBeats !== barBeats) throw new Error(`Bar
    ${index + 1}: ${melodyBeats} != ${barBeats}`);
  let position = startBeat;
  for (const note of bar.melody) {
    events.push({ track:"melody", pitch:note.pitch,
    startBeat:position, beats:note.beats });
    position += note.beats;
  }
  for (const pitch of bar.harmony) events.push({
    track:"chords", pitch, startBeat, beats:barBeats });
  for (let beat = 0; beat < barBeats; beat++) {
```

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11 / EVENTS

```
    events.push({ track:"bass", pitch:beat % 2 ?
    bar.bassFifth : bar.bassRoot, startBeat:startBeat +
    beat, beats:.8 });
  }
  startBeat += barBeats;
}
console.log(events);
```

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